Lecture 11: Infinite Series

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what an infinite series is and how it is formed from a sequence.
- ☐ Determine whether a series converges or diverges using partial sums.
- ☐ Recognize and work with geometric series, including finding their sums.
- ☐ Understand the nth-term test and apply it even if the question does not suggest it.
- ☐ Know how the manipulation of the series affecting its convergence/divergence behaviour.

Introduction

Consider the sequence $\{a_1, a_2, a_3, \ldots, a_n, \ldots\}$.

$S_n = a_1 + a_2 + \ldots + a_n$

Now, let define another sequence

$$\{s_n\} = \{\underbrace{a_1}_{s_1}, \underbrace{a_1 + a_2}_{s_2}, \underbrace{a_1 + a_2 + a_3}_{s_3}, \ldots, \underbrace{a_1 + a_2 + a_3 + \cdots + a_n}_{s_n}, \ldots\}$$

infinity.

As n grows larger, what is happening to this sequence $\{s_n\}$?

What is $\lim_{n\to\infty} s_n$? $\lim_{n\to\infty} s_n = \lim_{n\to\infty} a_1 + a_2 + \ldots + a_n$

$= a_1 + a_2 + \ldots + a_k + \ldots$

infinite series.

Infinite Series

Given a sequence of numbers $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

is an infinite series. The number a_n is called the nth term of the series. The sequence $\{s_n\}$ defined by

sequence of partial sum.

$$\begin{aligned} s_1 &= a_1 \\ s_2 &= a_1 + a_2 \\ &\vdots \\ s_n &= a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k \\ &\vdots \end{aligned}$$

partial sum.

is the sequence of partial sums of the series, and s_n is the nth partial sum.

Convergence and Divergence

If sequence $\{s_n\}$ converges to a number L, we say that the series converges and that its sum is L. In this case, we also write

$$a_1 + a_2 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n = L.$$

If $\{s_n\}$ does not converge, we say that the series diverges.

For simplicity, we also write $\sum_{n=1}^{\infty} a_n$ as $\sum a_n$.

Example

Determine whether $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}}$ converges or diverges, if it converges, evaluate the infinite series.

Let $a_n = \frac{1}{2^{n-1}}$, $\{a_n\}_{n=1}^{\infty} = \{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots\}$

$\{s_n\} = \{s_1, s_2, s_3, \dots, s_n, \dots\}$ where

$s_1 = 1 = 2-1$

$s_2 = 1+\frac{1}{2} = 2-\frac{1}{2}$

$s_3 = 1+\frac{1}{2}+\frac{1}{4} = 2-\frac{1}{4}$

$s_4 = 1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8} = 2-\frac{1}{8}$

$\vdots$

$s_n = 1+\frac{1}{1}+\dots+\frac{1}{2^{n-1}} = 2-\frac{1}{2^{n-1}}$

partial sum

nth partial sum.

So, the infinite series $1 + \frac{1}{2} + \frac{1}{4} + \cdots$

is $\lim_{n\to\infty} s_n = \lim_{n\to\infty} \left(2 - \frac{1}{2^{n-1}}\right) = 2.$

($\frac{1}{2^{n-1}} \to 0$)

infinite series : nth partial sum as $n \to \infty$.

Geometric Series

Geometric series are series of the form

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$$

where $a, r \in \mathbb{R}$ and $a \neq 0$.

A geometric series diverges if $|r| \geq 1$. ← $-1 \leq r \leq 1$

If $|r| < 1$, the geometric series converges and

$$\sum_{n=1}^{\infty} ar^{n-1} = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r} = \boxed{\frac{a}{1-r}}.$$

only valid when the geometric series converges.

If $r=1$, $S_n = a + a(1) + a(1)^2 + \dots + a(1)^{n-1}$

$= \underbrace{a + a + \dots + a}_{n \text{ times.}}$

$= na.$ $\Rightarrow \sum_{n=1}^{\infty} a(1)^{n-1}$ diverges.

$\to \infty$ as $n \to \infty$

If $r=-1$, $S_n = a + a(-1) + a(-1)^2 + \dots + a(-1)^{n-1}$

$= a - a + a - a + a - a + \dots + a(-1)^{n-1}$

$= \begin{cases} 0, & n \text{ is even} \\ a, & n \text{ is odd.} \end{cases}$

S_n is keep alternating between 0 and a

so $\lim_{n\to\infty} S_n = \sum_{n=1}^{\infty} a(1)^{n-1}$ does not exists (diverges)

For $r \neq 1$,

$$S_n = a + ar + ar^2 + \ldots + ar^{n-1} \quad \text{— ①}$$

$$rS_n = ar + ar^2 + ar^3 + \ldots + ar^n \quad \text{— ②}$$

① − ②, $S_n - rS_n = a - ar^n$

$$(1-r)S_n = a(1-r^n)$$

$$S_n = \frac{a(1-r^n)}{1-r}, \quad r \neq 1$$

When $|r|<1$, $r^n \to 0$ as $n \to \infty \Rightarrow S_n \to \frac{a}{1-r}$

When $|r|>1$, $r^n \to \infty$ as $n \to \infty \Rightarrow S_n$ diverges.

So $\lim_{n\to\infty} S_n = \begin{cases} \frac{a}{1-r}, & |r|<1 \\ \text{diverges}, & |r| \geq 1. \end{cases}$

Example

1. $a = \frac{1}{9}$ and $r = \frac{1}{3}$ $|r|<1$ $\sum_{n=1}^{\infty} ar^{n-1}$ converges.

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r} = \frac{1/9}{1-1/3} = 1/6$$

2. $\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n} = 5 - \frac{5}{4} + \frac{5}{4^2} - \frac{5}{4^3} + \cdots$

$a = 5$, $r = -\frac{1}{4}$ $\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n} = \frac{5}{1-(-1/4)} = \frac{5}{5/4}$

$(|r|<1)$

$= 4.$

Example (Telescoping Series)

Determine whether $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges or diverges, if it converges, evaluate the infinite series.

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \cdots$$

By the partial fraction decomposition,

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1} \quad \text{(Exercise)}$$

$$n=1: \frac{1}{1\cdot 2} = \frac{1}{1} - \frac{1}{2}$$

$$n=2: \frac{1}{2\cdot 3} = \frac{1}{2} - \frac{1}{3}$$

$$n=3: \frac{1}{3\cdot 4} = \frac{1}{3} - \frac{1}{4}$$

$$\vdots$$

When we add them up,

$$S_k = a_1 + a_2 + \cdots + a_k$$

$$\vdots$$

$n=k: \quad \frac{1}{k\cdot(k+1)} = \frac{1}{k} - \frac{1}{k+1}$ $\quad \rightarrow = 1 - \frac{1}{k+1}$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{n\to\infty} S_n$$

$$= \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right) \quad \frac{1}{n+1} \to 0$$

$$= 1$$

The nth-Term Test for Divergence

Considering the series

$$\sum_{n=1}^{\infty} \frac{n+1}{n} \quad \text{(diverges)}$$

$$\sum_{n=1}^{\infty} \frac{n+1}{n} = \frac{2}{1} + \frac{3}{2} + \frac{4}{3} + \cdots$$

$$> 1 + 1 + 1 + \cdots$$

What happens to S_n?

$$S_n = \frac{2}{1} + \frac{3}{2} + \frac{4}{3} + \cdots + \frac{n+1}{n}$$

$$> \underbrace{1 + 1 + \cdots + 1}_{n \text{ terms}} = n$$

$S_n > n \Rightarrow$ As $n \to \infty$, $S_n \to \infty$

The nth-Term Test for Divergence

Theorem: If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \to 0$.

$\left(\lim_{n\to\infty} a_n = 0\right)$

Proof: Let $S_n = \sum_{n=1}^{\infty} a_n$

Suppose that $\sum_{n=1}^{\infty} a_n$ converges to $S \in \mathbb{R}$, then $\lim_{n\to\infty} S_n = S$

the nth term partial sum $\therefore S_n = \underbrace{a_1 + a_2 + \dots + a_{n-1}}_{S_{n-1}} + a_n$

$$= S_{n-1} + a_n$$

$$a_n = S_n - S_{n-1}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (S_n - S_{n-1}) = \lim_{n\to\infty} S_n - \lim_{n\to\infty} S_{n-1} = S - S = 0$$

General idea / General picture.

$$a_1 + a_2 + \cdots + a_{n-1} + a_n + a_{n+1} + \cdots \not\to 0$$

this means we will always have something to add.

So the sum cannot be convergent.

The nth-Term Test for Divergence

The nth-Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n\to\infty} a_n$ fails to exist or is different to zero.

If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Contrapositive of previous theorem. (slide 11/19)

Caution: $\lim_{n\to\infty} a_n = 0$ or fails to exists does not necessary mean that $\sum_{n\to\infty}^{\infty} a_n$ converges.

The nth-Term Test for Divergence

Examples:

1. $\sum_{n=1}^{\infty} n^2$ $\quad a_n = n^2 \to \infty$ (the series diverges)

2. $\sum_{n=1}^{\infty} \frac{n+1}{n}$ $\quad a_n = \frac{n+1}{n} = 1 + \frac{1}{n} \to 1 \neq 0$ (the series diverges)

3. $\sum_{n=1}^{\infty} (-1)^{n+1}$ $\quad a_n = (-1)^{n+1} = \begin{cases} 1, & \text{if } n \text{ is odd} \\ -1, & \text{if } n \text{ is even} \end{cases}$ $\Rightarrow$ It keeps alternating between -1 and 1, so the limit does not exist. (the series diverges)

4. $\sum_{n=1}^{\infty} \frac{-n}{2n+5}$

④ $a_n = \frac{-n}{2n+5} \longrightarrow \frac{-1}{2} \neq 0$ (the series diverges)

$s_n = a_1 + a_2 + \dots + a_n$

$\lim_{n\to\infty} s_n = a_1 + a_2 + \dots + a_k + \dots$

$= a_1 + a_2 + \dots + a_k + \dots + \left(\frac{-1}{2}\right) + \left(\frac{-1}{2}\right) + \dots$

↓

As $n \to \infty$, we will keep adding the infinitely many number of $-\frac{1}{2}$, the addition never stops, so the series diverges.

(As long as $a_n \to S \neq 0$, the addition never stop, the series diverges.)

The nth-Term Test for Divergence

Example: Consider the series

$$1+\underbrace{\frac{1}{2}+\frac{1}{2}}_{2\text{ times}}+\underbrace{\frac{1}{4}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}}_{4\text{ times}}+\cdots+\underbrace{\frac{1}{2^n}+\frac{1}{2^n}+\cdots+\frac{1}{2^n}}_{2^n\text{ times}}+\cdots$$

$=1$ $=1$ $=1$

Let define our partial sum as follows.

$s_1 : 1 \geq 1$

$s_2 : 1+\frac{1}{2}+\frac{1}{2} \geq 2$

$s_3 : 1+\frac{1}{2}+\frac{1}{2}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4} \geq 3$

$\vdots$

Inductively, the series $s_k \geq k$ for any k (the series diverges)

Eventhough the sequence $a_n = \left\{\frac{1}{2^n}\right\}$ converges to 0.

This means the converse of nth term test for divergent is not necessarily true.

($\{a_n\} \to 0$ does not tell us anything about the convergency of $\sum a_n$)

Combining Series

Theorem: If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. Sum Rule: $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. Difference Rule: $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. Constant Multiple Rule: $\sum ka_n = k\sum a_n = kA$ for any number k

Let $A_n = a_1 + a_2 + a_3 + \cdots + a_n \longrightarrow A$ } nth term partial sum.

$B_n = b_1 + b_2 + \cdots + b_n \longrightarrow B.$ }

Let S_n be the nth partial sum of $\sum (a_n + b_n)$

$$S_n = (a_1 + b_1) + (a_2 + b_2) + \cdots + (a_n + b_n)$$

$$= (a_1 + a_2 + \cdots + a_n) + (b_1 + b_2 + \cdots + b_n)$$

$$= A_n + B_n \longrightarrow A + B.$$

Let t_n be the nth partial sum of $\Sigma(ka_n)$

$$t_n = ka_1 + ka_2 + \dots + ka_n$$

$$= k(a_1 + a_2 + \dots + a_n)$$

$$= kA_n \longrightarrow kA.$$

Combining Series

As corollaries of previous theorem: (contrapositive of theorem > slide 15/19)

1. Every nonzero constant multiple of a divergent series diverges.
2. If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ and $\sum(a_n - b_n)$ both diverge.

(or Σa_n diverges and Σb_n converges)

Be aware!!

It is not necessarily true that

if $\Sigma(a_n + b_n)$ converges, then Σa_n & Σb_n converges.

Eg: $a_n = (-1)^n$, $b_n = (-1)^{n+1}$

Σa_n diverges Σb_n diverges.

$\Sigma(a_n + b_n) = 0$ (converges)

Combining Series

Examples:

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}} = \sum_{n=1}^{\infty}\left(\frac{3^{n-1}}{6^{n-1}} - \frac{1}{6^{n-1}}\right)$

$= \dfrac{3^{n-1}}{6^{n-1}} - \dfrac{1}{6^{n-1}} = \displaystyle\sum_{n=1}^{\infty}\left(\left(\frac{1}{2}\right)^{n-1} - \left(\frac{1}{6}\right)^{n-1}\right)$

$|r|<1$

geometric series $a=1, r=\frac{1}{2}$ converges to $\dfrac{1}{1-\frac{1}{2}}$

geometric series $a=1, r=\frac{1}{6}$ converges to $\dfrac{1}{1-\frac{1}{6}}$

$= \dfrac{1}{1-1/2} - \dfrac{1}{1-1/6} = 4/5$

(b) $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{4}{2^n}$

(b) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{4}{2^n} \times 2 = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{4 \cdot 2}{2^n \cdot 2}$

$= \sum_{n=1}^{\infty} 8 \left(\frac{-1}{2}\right)^{n+1}$

$= 8 \sum_{n=1}^{\infty} \left(\frac{-1}{2}\right)^{n+1}$ geometric series with.

$a = \frac{1}{4}$, $r = \frac{-1}{2}$

$n=1, \left(\frac{-1}{2}\right)^2$

$n=2, \left(\frac{-1}{2}\right)^3 = \left(\frac{-1}{2}\right)^2 \cdot \left(\frac{-1}{2}\right)$

$= 8 \left[\frac{1/4}{1-(-1/2)}\right]$

$= 4/3$

Adding or Deleting Terms

Adding or deleting finite terms to or from a series will not affect the series's convergence or divergence. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$.

Conversely, if $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$, then $\sum_{n=1}^{\infty} a_n$ converges.

* But of course, the sum might be different

Eg: $\sum_{n=1}^{\infty} \frac{1}{5^n}$ converges $\Rightarrow$ $\sum_{n=10000000}^{\infty} \frac{1}{5^n}$ converges.

$\sum_{n=1}^{\infty} \frac{1}{n}$ diverges $\Rightarrow$ $\sum_{n=10000000}^{\infty} \frac{1}{n}$ diverges.

(harmonic series)

Reindexing

As long as the order of its terms is preserved, reindexing any series will not alter its convergence.

Eg: We can write

$$\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 + \frac{1}{2} + \frac{1}{4} + \dots$$

as $\sum_{k=0}^{\infty} \frac{1}{2^k}$ or $\sum_{k=5}^{\infty} \frac{1}{2^{k-5}}$ or $\sum_{k=-1}^{\infty} \frac{1}{2^{k+1}}$

(Let $k = n-1$)

(They are all reflecting to a same series)